

Statistical Data Analysis

Lecture 13: Linear regression analysis, part 3

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Periods 4 & 5

First announcement

All material shared at macostanje.github.io/sda

Where Are We?

◀ Chapter 7: Analysis of categorical data

💬 Still today: discussion of Assignment 6 & Quiz 6

📍 Chapter 8: Linear regression analysis

◀ Section 8.1 – The multiple linear regression model

◀ Section 8.2 – Diagnostics

➤ Section 8.3 – Collinearity

💬 Still on May 20th: discussion of Assignment 7 & Quiz 7

Recap – multiple linear regression

Recap – multiple linear regression

- › Linear regression equation (scalar)

$$Y_i = \beta_0 + x_{i1}\beta_1 + \cdots + x_{ip}\beta_p + e_i$$

- › Linear regression equation (vectorized)

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$

- › Assumptions on errors

$$\mathbb{E}(\mathbf{e}_i) = 0 \quad \mathbb{E}(\mathbf{e}_i \mathbf{e}_j) = \begin{cases} \sigma^2, & i = j, \\ 0, & i \neq j \end{cases}$$

- › Further **assumptions**

- › Full-rank **design matrix**, i.e. $\text{rank}(\mathbf{X}) = p + 1$
- › $e_i \sim N(0, \sigma^2)$, so that $Y \mid \mathbf{X} = \mathbf{x} \sim N(\beta_0 + x_1\beta_1 + \cdots + x_p\beta_p, \sigma^2)$

Collinearity

Collinearity – Definition

- Two explanatory variables X_1 and X_2 are called (perfectly) **collinear** if there is a (exact) linear relationship between X_1 and X_2
 - If more than two predictors are linearly related, we sometimes speak of **multicollinearity**
- Perfect collinearity:** $X_2 = c_0 + c_1 X_1$
 - Example: X_1 (height in m) and $X_2 = 100X_1$ (height in cm)
 - Intuition:** no variation in response can be explained by X_2 if X_1 is in the model
 - Issues:** X does not have full rank, $X^T X$ **not** invertible, coefficient estimates $\hat{\beta}$ **cannot** be determined
- Strong, but not exact, linear relationship $\rightsquigarrow X_1, X_2$ (approximately) **collinear**
 - Intuition:** the above is **almost** true
 - Issues:** numerical issues in computing $(X^T X)^{-1}$, large estimator variances

Collinearity – Illustration

```
library(ISLR)
summary(lm(Balance ~ Limit + Age, data = Credit))$coefficients
```

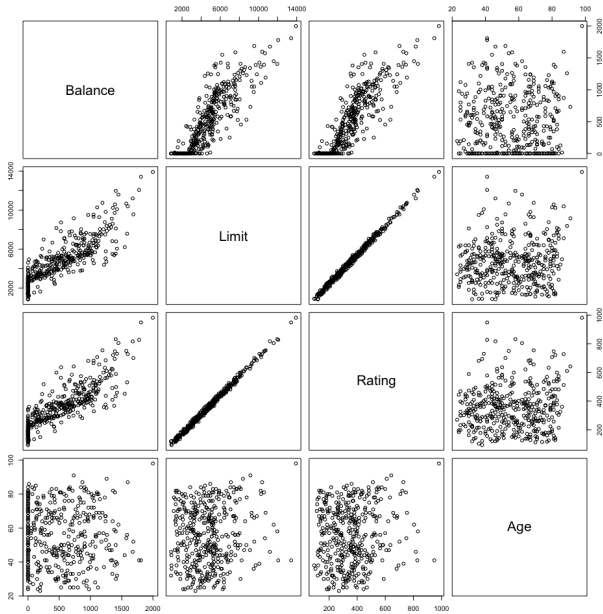
##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-173.410901	43.828387048	-3.956589	9.005366e-05
## Limit	0.173365	0.005025662	34.495944	1.627198e-121
## Age	-2.291486	0.672484540	-3.407492	7.226468e-04

```
summary(lm(Balance ~ Limit + Rating + Age, data = Credit))$coefficients
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-259.51751854	55.88219241	-4.6440110	4.656409e-06
## Limit	0.01901346	0.06296388	0.3019741	7.628303e-01
## Rating	2.31045954	0.93952549	2.4591771	1.435238e-02
## Age	-2.34575164	0.66861406	-3.5083792	5.025853e-04

- ❓ What happened to standard errors and significance results for Limit after the inclusion of Rating?

Collinearity – Illustration



Another Example – Seat Position

- › Dataset `seatpos` in package `faraway` contains data about 38 car drivers
- › From the documentation:
 - › `Age` denotes age (in years)
 - › `Weight` denotes weight (in lbs)
 - › `HtShoes` denotes height when wearing shoes (in cm)
 - › `Ht` denotes height when barefoot (in cm)
 - › `Seated` denotes height when seated (in cm)
 - › `Arm` denotes lower arm length (in cm)
 - › `Thigh` denotes thigh length (in cm)
 - › `Leg` denotes lower leg length (in mm)
 - › `hipcenter` denotes horizontal distance of the midpoint of the hips from a fixed location in the car (in cm)

Collinearity – Seat Position Example

```
library(faraway)
# Note: ~ . denotes regression on all other variables in the data frame
summary(lm(hipcenter ~ ., data = seatpos))

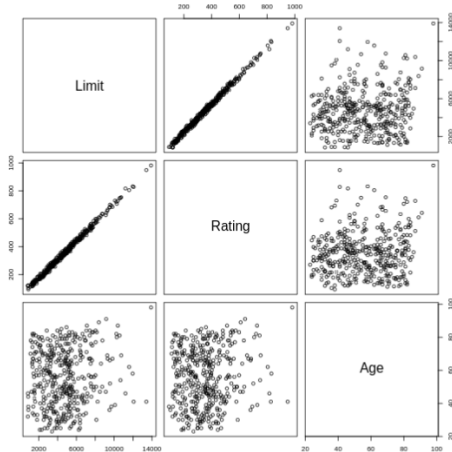
##
## Call:
## lm(formula = hipcenter ~ ., data = seatpos)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -73.827 -22.833  -3.678  25.017  62.337
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  436.43213   166.57162    2.620   0.0138 *
## Age           0.77572     0.57033    1.360   0.1843
## Weight        0.02631     0.33097    0.080   0.9372
## HtShoes       -2.69241     9.75304   -0.276   0.7845
## Ht            0.60134    10.12987    0.059   0.9531
## Seated        0.53375     3.76189    0.142   0.8882
## Arm          -1.32807     3.90020   -0.341   0.7359
## Thigh         -1.14312     2.66002   -0.430   0.6706
## Leg          -6.43905     4.71386   -1.366   0.1824
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 37.72 on 29 degrees of freedom
## Multiple R-squared:  0.6866, Adjusted R-squared:  0.6001
## F-statistic: 7.94 on 8 and 29 DF,  p-value: 1.306e-05
```

- ② What conclusion do you draw from the overall F -test?
- ② What about the individual t -tests?

How to Investigate (Potential) Collinearity?

- › Pairwise scatterplots & pairwise correlations among predictors
 - › First steps for graphical resp. numerical investigation of pairwise collinearity
- › Variance inflation factors VIF_j of $\hat{\beta}_j, j = 1, \dots, p$
 - › Which estimate $\hat{\beta}_j$ cannot be trusted (if any)?
- › Condition number $\kappa(X)$ of design matrix X
 - › Is there collinearity?
- › Condition indices $\kappa_k(X), k = 1, \dots, p + 1$
 - › How many collinearities?
- › Variance decomposition of $\hat{\beta}_j, j = 1, \dots, p$
 - › Which $X_k, k \neq j$ are involved in a collinearity?

Pairwise Scatterplots & Correlations Among Predictors



```
cor(Credit[, c(3, 4, 6)])
```

```
##           Limit    Rating    Age
## Limit  1.0000000 0.9968797 0.1008879
## Rating 0.9968797 1.0000000 0.1031650
## Age    0.1008879 0.1031650 1.0000000
```

Measures of Collinearity

Measures of Collinearity – Variance Inflation Factors

- › **Question:** which estimate $\hat{\beta}_j$ cannot be trusted (if any)?
- › **Idea:** regress X_j on all other predictors
 - › Denote the corresponding determination coefficient by $\mathcal{R}_j^2$
 - › Very **high** $\mathcal{R}_j^2 \rightsquigarrow X_j$ **almost linear combination** of other predictors
- › Quantify this with **variance inflation factor** $VIF_j = \frac{1}{1-\mathcal{R}_j^2}$, $j = 1, \dots, p$
 - › VIF_j **much larger** than 1 $\rightsquigarrow \hat{\beta}_j$ **unreliable**
 - › **Rule-of-thumb:** $VIF_j > 5$ for some j points to **collinearity** in the model
 - › Still-open question: which variables are involved in which collinear relationships?

```
library(car)
vif(lm(Balance ~ Limit + Rating + Age, data = Credit))
```

```
##      Limit      Rating      Age
## 160.592880 160.668301  1.011385
```

Variance Inflation Factors – Seat Position Example

> Question: which estimate $\hat{\beta}_j$ cannot be trusted (if any)?

```
car::vif(lm(hipcenter ~ ., data = seatpos))
```

##	Age	Weight	HtShoes	Ht	Seated	Arm	Thigh
##	1.997931	3.647030	307.429378	333.137832	8.951054	4.496368	2.762886
##	Leg						
##	6.694291						

Measures of Collinearity – Condition Number

- › Question: is there collinearity?
- › Idea: **perfectly collinear** columns in $X \rightsquigarrow \text{rank}(X) < p + 1$
 - › At least one **eigenvalue** ν_j of $X^T X$ zero
 - › Coefficient estimates $\hat{\beta}$ **not computable**
 - › $\min \nu_j \approx 0$ instead? $\rightsquigarrow$ large estimator variance(s)
- › Quantify this with **condition number** of X : $\kappa(X) = \sqrt{\frac{\max_j \nu_j}{\min_j \nu_j}}$
 - › $\kappa(X)$ large $\rightsquigarrow X$ is **ill-conditioned**
 - › $\kappa(X) > 30 \rightsquigarrow$ further **investigation** of collinearity is necessary
 - › Note: eigenvalues depend on the scaling of columns, rule-of-thumb valid if columns have similar scales
 - › Columns of X can be scaled to e.g. have Euclidean length of 1: this is useful in diagnostics, but meaning can get lost!

```
library(multiColl)
X <- model.matrix(lm(Balance ~ Limit + Rating + Age, data = Credit))
# Scale columns (leave intercept untouched)
X[, -1] <- apply(X[, -1], 2, function(x) x/sqrt(sum(x^2)))
CN(X)
```

```
## [1] 83.52181
```

Condition Number – Seat Position Example

> Question: is there collinearity?

```
X_seatpos <- model.matrix(lm(hipcenter ~ ., data = seatpos))  
# Scale columns (leave intercept untouched)  
X_seatpos[, -1] <- apply(X_seatpos[, -1], 2, function(x) x/sqrt(sum(x^2)))  
multiColl::CN(X_seatpos)  
  
## [1] 1153.483
```

Measures of Collinearity – Condition Indices

› Question: how many collinear relationships are there?

› Condition indices of X : $\kappa_k(X) = \sqrt{\frac{\max_j \nu_j}{\nu_k}}$

› Here written in terms of ν 's

› $\kappa_k(X) \geq 30 \rightsquigarrow$ indication of a collinearity

› Note: again, eigenvalues depend on the scaling of columns

```
library(olsrr)
round(ols_eigen_cindex(lm(Balance ~ Limit + Rating + Age, data = Credit))[, 1:2], 3)
```

##	Eigenvalue	Condition Index
## 1	3.765	1.000
## 2	0.192	4.427
## 3	0.042	9.457
## 4	0.001	83.522

Condition Indices – Seat Position Example

> **Question:** how many collinear relationships are there?

```
round(olsrr::ols_eigen_cindex(lm(hipcenter ~ ., data = seatpos))[, 1:2], 2)
```

##	Eigenvalue	Condition Index
## 1	8.82	1.00
## 2	0.14	7.83
## 3	0.03	17.20
## 4	0.00	43.52
## 5	0.00	55.58
## 6	0.00	79.52
## 7	0.00	116.37
## 8	0.00	213.60
## 9	0.00	1153.48

Measures of Collinearity – Variance Decomposition

- › Singular value decomposition of design matrix X into $X = UDV^T$
 - › U : full-rank $n \times (p + 1)$ matrix
 - › D : diagonal $(p + 1) \times (p + 1)$ matrix
 - › V : full-rank $(p + 1) \times (p + 1)$ matrix
 - › $U^T U = V^T V = I$
- › Singular values of X , $\mu_1, \dots, \mu_{p+1}$: diagonal entries of D
 - › Their squares equal the eigenvalues $\nu_1, \dots, \nu_{p+1}$ of $X^T X$
- › Write the estimator variance as $\text{Var}(\hat{\beta}_j) = \sigma^2 \sum_{k=1}^{p+1} \frac{v_{jk}^2}{\mu_k^2}$
 - › Each component involves one among $\mu_1, \dots, \mu_{p+1}$
 - › Contribution is large if μ_k is small $\iff \kappa(X)_k$ large
- › Variance decomposition proportions of $\hat{\beta}_j$: relative contribution π_{kj} of each term in the sum

$$\pi_{jk} = \frac{v_{jk}^2 / \mu_k^2}{\sum_k v_{jk}^2 / \mu_k^2}$$

Variance Decomposition Contributions

- **Question:** which predictors are involved in a collinear relationship?

```
library(olsrr)
round(ols_eigen_cindex(lm(Balance ~ Limit + Rating + Age, data = Credit)), 3)
```

##	Eigenvalue	Condition	Index	intercept	Limit	Rating	Age
## 1	3.765		1.000	0.003	0.000	0.000	0.006
## 2	0.192		4.427	0.036	0.002	0.001	0.157
## 3	0.042		9.457	0.579	0.000	0.000	0.837
## 4	0.001		83.522	0.382	0.998	0.999	0.001

- Column Condition Index: $\kappa_k(X)$
- Other columns (note that of them each sums to 1): relative contributions to variance of $\hat{\beta}_j$ for $j = 0, 1, 2, 3$
- **Interpretation:** Limit and Rating involved in a collinearity

Variance Decomposition Contributions – Seat Position Example

➤ Question: which predictors are involved in a collinear relationship?

```
# Removed Eigenvalue column for space reasons
```

```
round(olsrr::ols_eigen_cindex(lm(hipcenter ~ ., data = seatpos))[, -1], 2)
```

	Condition Index	intercept	Age	Weight	HtShoes	Ht	Seated	Arm	Thigh	Leg
1	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	7.83	0.00	0.48	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3	17.20	0.01	0.00	0.35	0.00	0.00	0.00	0.00	0.00	0.00
4	43.52	0.05	0.02	0.08	0.00	0.00	0.01	0.04	0.46	0.00
5	55.58	0.03	0.05	0.19	0.00	0.00	0.00	0.40	0.29	0.07
6	79.52	0.09	0.26	0.12	0.00	0.00	0.00	0.51	0.00	0.51
7	116.37	0.80	0.11	0.24	0.00	0.00	0.13	0.00	0.05	0.23
8	213.60	0.02	0.00	0.00	0.02	0.01	0.86	0.03	0.12	0.19
9	1153.48	0.00	0.08	0.01	0.98	0.98	0.00	0.01	0.08	0.00

Collinearity Remedies (?)

- › No standard fix to solve a collinearity issue
- › Possible remedies to try may include
 - › Investigation of outliers/leverage points
 - › Deletion/replacement of columns
 - › ...

Summary & wrap-up

› Collinearity

- › If two or more explanatory variables are linearly dependent, estimating coefficients becomes problematic
- › Results become unreliable

› Investigating collinearity

- › Pairwise scatterplots and correlation matrix
- › Variance inflation factors VIF_j
- › Condition number $\kappa(X)$
- › Condition indices $\kappa_k(X)$
- › Variance decomposition of $\hat{\beta}_j$

What now?

- › To do for now
 - › Continue assignment 7
 - › **Deadline:** May 19 at 23:59
 - › If Canvas still not functional by then, follow instructions sent over e-mail
- › After that
 - › No lectures
 - › Next week: discussion of old exam (see Canvas or [web page](#))